

Charles T. Marx

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EDUCATION:

B. Sc. Candidate in Computer Science and Mathematics, Minor in Statistics
Haverford College (GPA 3.94/4.00)

PRIMARY RESEARCH INTERESTS:

Machine learning, causal inference and applications of these techniques to topics such as model interpretability, fairness and environmental science.

POSITIONS:

Research Intern (2019) – Harvard University

Developed methods for finding equivalent models for recidivism prediction and theoretical work to develop their implications. Advised by Burk Ustun and Flavio Calmon.

Research Assistant (2017 – present) – Dept. of Computer Science, Haverford College

Machine learning research spanning adversarial learning paradigms, disentangled representations, and fairness and interpretability of black box models. Advised by Sorelle Friedler.

Research Assistant (2017 – 2018) – Dept. of Chemistry, Haverford College

Data mining and machine learning in environmental chemistry. Published work using machine learning to assist in rapid identification of oil in the environment. Advised by Helen White.

Research Assistant (2018) – Digital Scholarship, Haverford College

Used recurrent neural networks for handwriting recognition in a dataset of 100,000 handwritten Spanish texts.

Summer Research Intern (2018) – Prince William Sound Science Center, Alaska

Data collection and analysis in oceanography, such as aerial herring surveys and applications of machine learning to interpret sonar data. Advised by Scott Pegau.

AWARDS:

Beckman Scholars Award (2018)

National undergraduate scholarship for excellence in research.

Farmers Insurance Standout Student of the Year Award (2017)

Regional scholarship for academic excellence and community engagement awarded to one student each year.

Brian Kovaric Fund Finalist (2017)

Faculty-nominated scholarship for one promising first-year student.

Travel Grants

- Summer Cluster on Fairness – Simon’s Institute for the Theory of Computing, Berkeley, CA (‘19)
- SIGKDD - Special Interest Group on Knowledge Data Discovery & Data Mining Conference, London, UK (‘18)
- FATML - Fairness, Accountability and Transparency in Machine Learning Conference, NY, NY (‘18, ‘19)
- GOMOSEs - Gulf of Mexico Oil Spill Research Conference, New Orleans, LA (‘17)
- Functional Ecological Genomics Workshop, Lacawac Sanctuary & Biological Field Station (‘17)

PUBLICATIONS

1. A.P.M. Michel, A.E. Morrison, **C.T. Marx**, H.K. White (2018) Rapid Identification of Deepwater Horizon Oil Residues using X-Ray Fluorescence. Environmental Science & Technology Letters (Accepted).
2. H.K. White, **C.T. Marx**, D.L. Valentine, C. Sharpless, C. Aeppli, K.M. Gosselin, V. Kivenson, R.M. Liu, R.K. Nelson, S.P. Sylva, C.M. Reddy (2019) Examining inputs of biogenic and oil-derived hydrocarbons in surface waters following the Deepwater Horizon incident. ACS Earth & Space Chemistry (Accepted).
3. **Charles T. Marx**, Sorelle A. Friedler, Carlos Scheidegger, Suresh Venkatasubramanian (2019) Disentangling Influence: Using disentangled representations to audit model predictions. NeurIPS (In review).

ORAL PRESENTATIONS:

1. **C. T. Marx**, D. L. Valentine, C. M. Reddy, H. K. White (2018) Examining Inputs of Biogenic and Oil-Derived Hydrocarbons in Surface Waters Following the Deepwater Horizon Oil Spill, GOMOSEs Conference, New Orleans, Louisiana.
2. Scheidegger, C., S. Venkatasubramanian, **C. T. Marx** (2018) Auditing Black Box Models. Fairness, Accountability and Transparency in Machine Learning (FAT*) Conference, New York, New York.
3. Reddy, C.M., K.M. Gosselin, R.K. Nelson, S. P. Sylva, C.Aeppli, C.M. Sharpless, D.L. Valentine, **C.T. Marx**, H.K. White (2017) A Biogenic Input of Heptadecane in Surface Waters Prior to and during the Anthropocene. Goldschmidt Conference, Paris, France.

POSTER PRESENTATIONS:

1. **C.T. Marx**, S.A. Friedler, H.K. White (2018) Deep Learning Methods for Minimizing Social Discrimination. Regional Beckman Symposium, NY, NY.
2. **C.T. Marx**, C. M. Reddy, D.L. Valentine, H.K. White (2017) Examining Biogenic and Oil-Derived Hydrocarbons in Surface Waters After the Deepwater Horizon Oil Spill. Marian E. Koshland Integrated Natural Sciences Center Undergraduate Research Symposium, Haverford, PA.

FIELD EXPERIENCE:

Pensacola, FL to Grand Isle, LA (06/19/17-06/23/17)

Collection of oil from Gulf of Mexico beaches impacted by the Deepwater Horizon oil spill.

Auklet Charter: Prince William Sound, Alaska (07/25/18-07/31/18)

Collection of oil residues from shorelines impacted by the 1989 Exxon Valdez oil spill and the 1964 Alaska Earthquake.

ACADEMIC SERVICE:

Computer Science Faculty Search Committee (2018-19)

Member of group tasked with hiring a tenure-track faculty member for the computer science department.

Perovskite Solar Cell Chemistry Group (2017)

An extracurricular chemistry lab to design perovskite solar cells for the introductory lab curriculum.

EXTRACURRICULAR SERVICE:

Furniture Forward (2015-2017)

Founder of organization to redistribute furniture to families transitioning out of homelessness.

Outdoors Club Leader (2017-2018)

Designed and led week-long backpacking trips for first-year students at Haverford College.

Haverford College Men's Club Soccer (2018)

Captain.